

Nathan Xia

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Engineering Portfolio: Nathan Xia

EDUCATION

- **University of Toronto** Toronto, Canada
Bachelor of Applied Science - Engineering Science | Lachlan Dales McKeller Scholar *September 2025 - Present*
cGPA: 3.95/4.00

SKILLS SUMMARY

- **Languages:** Python, Java, MATLAB, C, C++ (Arduino)
- **Tools:** ANSYS Fluent, Autodesk CFD, SolidWorks, Fusion 360, Autodesk Inventor, OpenRocket, OpenVSP
- **Skills:** CFD simulation and meshing, aerodynamic analysis and design, CAD/CAM, wind tunnel analysis and design, signal processing and analysis

EXPERIENCE

- **U of T Institute for Aerospace Studies** FCET Research Group
Research Intern *May 2026 - September 2026*
 - Selected as **1 of only 2 first-year students** to conduct research at UTIAS, independently leading an aeroacoustics study on remote microphone probe calibration for wind-tunnel pressure measurements. Found here: Preprint
 - Owned the full experimental workflow from apparatus and test-matrix design through **data acquisition and FFT-based signal processing** across 16 investigated variables, producing a **70+ GB dataset across 1,200+ collected data files**.
 - Developed a **MATLAB** acquisition and analysis pipeline and identified improvements that **reduced transfer function magnitude standard deviation by 69%** and **extended the usable frequency range to 1.5×** its original range.
 - Experimentally validated findings through **20+ trials** that informed the group's ongoing RMP research and future RMP projects, including improved calibration configurations and procedures.
- **Robotics for Space Exploration - Aerial** University of Toronto
Mechanical Co-Lead *July 2026 - Present*
 - Direct all mechanical design and integration activities for the subsystem, with responsibility for architecture, **CAD**, analysis, manufacturing, assembly, and final system integration.
 - Lead a team of mechanical recruits, overseeing **CAD onboarding, CFD analysis**, design iteration, and manufacturing while reviewing and coordinating technical work across the team.
- *Mechanical Sub-team Recruit* *September 2025 - July 2026*
 - Conceived the glider's lifting-body configuration, replacing a conventional fuselage with an airfoil-profile body; evaluated **20+ airfoils in OpenVSP/VSPAERO** to select the final geometry.
 - Integrated passive flow-control devices, including vortex generators and end plates, to mitigate boundary-layer separation, increasing simulated lift by **5× at 20 m/s**.
 - Designed and simulated rocket nose-cone geometries in Autodesk CFD, iterating through **30+ profiles** using CFD and rocket-performance simulations.
- **STEM Fellowship REO** University of Calgary
Student Lab Intern *March 2025*
 - Selected as **1 of 30** from a competitive pool of ≈ 150 **applicants** for a research internship, collaborating with graduate researchers and professors on carbon-cycle and seawater CO₂ removal.
 - Performed chemical sample analysis using ion chromatography (IC), optical emission spectroscopy (OES), and scanning electron microscopy (SEM), and presented findings to ≈ 60 **researchers** at the University of Calgary.

PROJECTS

- **Lattice Boltzmann CFD Solver:** This project is ongoing. View Results.
Developed and verified a Python-based D2Q9 Lattice Boltzmann CFD solver from scratch featuring configurable geometries and flow conditions, velocity, pressure, and vorticity visualization, and aerodynamic force tracking. Benchmarked cylinder-flow behaviour at $Re = 150$ yielding a C_d **within 1.4% of a published result, reproducing von Kármán vortex shedding**, and the expected oscillations in lift and drag forces. Further benchmarked the solver on a NACA 0012 at $Re = 2000$ and $AOA = 8^\circ$, **producing a Strouhal number within 1.8% of a published ANSYS Fluent result**.
- **Vortex Generator Research Project: Paper**
Designed and conducted a vortex-generator study on a NACA 2412 airfoil using a self-designed and manufactured flow-visualization wind tunnel. Identified an optimal gap width and chordwise placement; regression predicted a **76% increase in the angle at which separation reached 30% chord**. Designed the airfoil in Desmos and CAD and presented the findings at CYSF and CWSF. Selected as **1 of 15 projects from ≈ 600** to compete at CWSF and earned a bronze medal among ≈ 350 projects.
- **ANSYS Fluent Analysis — Vortex Generator Project Extension:** View Details.
Extended the study with **3D steady laminar ANSYS Fluent simulations** at $Re \approx 4,300$, reproducing the wind-tunnel geometry and no-slip boundaries. Built locally refined polyhedral meshes with boundary-layer inflation and verified convergence using residuals, force histories, and mass balance. A **two-mesh sensitivity study** produced differences of **0.9% in lift coefficient, 1.4% in drag coefficient, and 1.3 mm in separation location**. Compared control and VG separation using signed wall shear stress and identified force-coefficient sensitivity to domain confinement.